TASK SPECIFICATION: P7-04 PHASE P7 — NOTIFICATIONS, GOVERNANCE & TELEMETRY ANALYTICS

System Administration & Governance API

Git Branch: `feature/P7-04-admin-api` · Target Pull Request: feature/P7-04-admin-api

Core Objective & Technical Scope:
Privileged administrative endpoints for managing campus parking spots, study rooms, tariff rules, and user role assignments.

Priority Level	Medium	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`feature/P7-04-admin-api`
Upstream Dependencies	P5-01, P6-06		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Implement admin controllers with @PreAuthorize("hasRole('CAMPUS_ADMIN')").
2. Provide administrative CRUD endpoints for spots, rooms, and tariffs.
3. Record all administrative overrides in audit log ledger.
4. Write security tests verifying access restriction.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `Admin endpoints across domain modules`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Non-admin requests receive 403 Forbidden with RFC 7807 problem details.
- Every administrative modification produces an audit trail.

VERIFICATION & TEST COVERAGE STRATEGY

- AdminControllerTest
- SecurityMatrixTest
- AuditLoggingTest